Proof of Concept: SafetyScanAI

Real-Time AI-Powered Safety Monitoring & Compliance System

Executive Summary:

SafetyScanAI is a state-of-the-art diagnostic and monitoring system designed to automate industrial safety inspections. By integrating computer vision, Voice AI (speech to text), retrieval-augmented generation (RAG), and multi-agent orchestration, the system detects safety violations (e.g., PPE non-compliance), retrieves relevant legal consequences (OSHA standards), and generates automated compliance reports emails. This POC demonstrates a hybrid-cloud architecture that bridges local data acquisition with high-performance AI processing.

The Challenges:

Manual safety monitoring in industrial environments is:

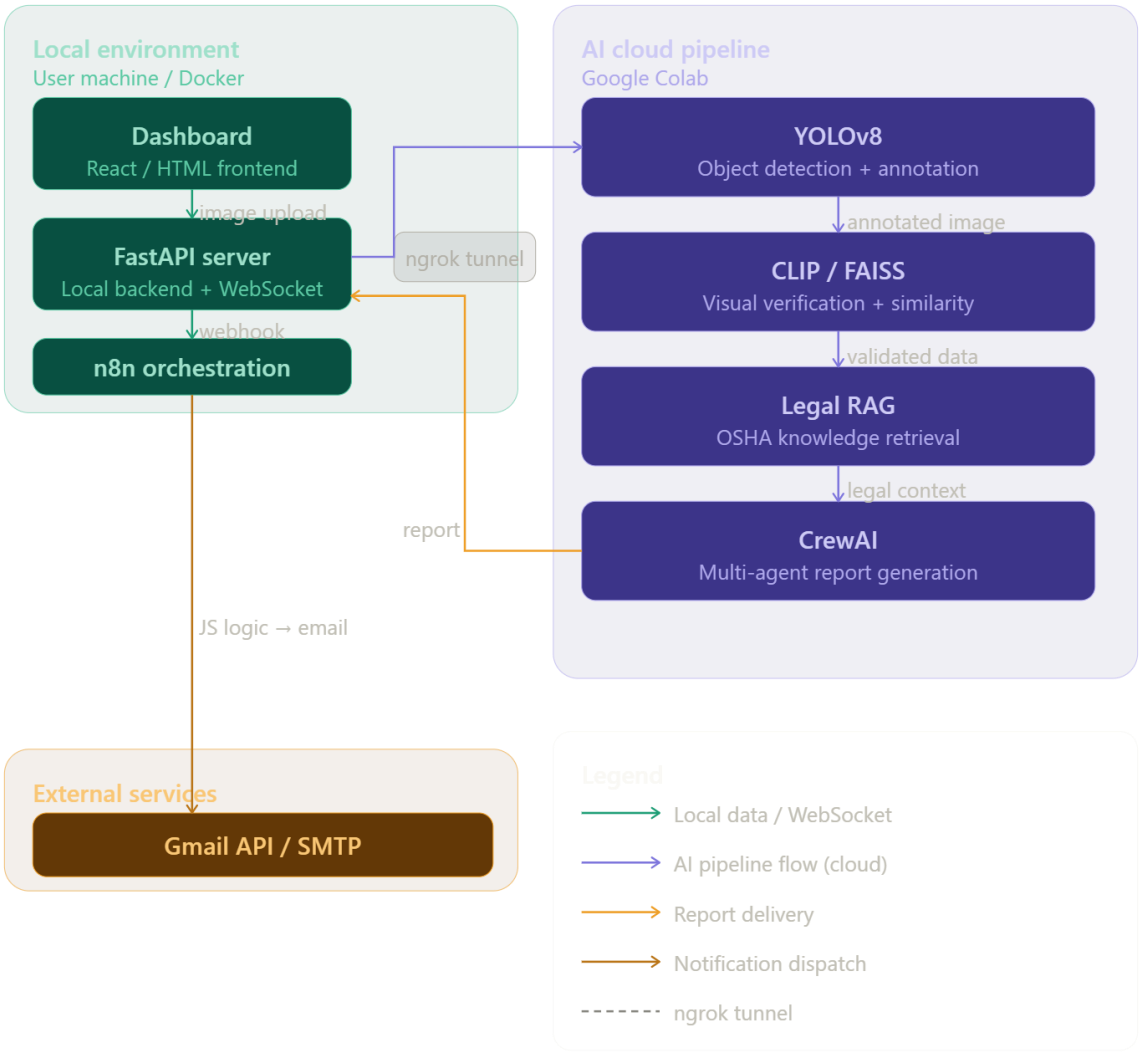
- **Time-Consuming**: Requires 24/7 human oversight.
- **Error-Prone**: Human fatigue leads to missed violations.
- **Reactive**: Compliance reports are often filed days after an incident.
- **Scaling Difficulty**: Traditional systems lack the reasoning capability to explain *why* a violation occurred or what the legal implications are.

The Solution: SafetyScanAI

SafetyScanAI provides a fully automated, end-to-end pipeline:

1. **Vision Detection**: Identifies PPE violations in real-time.
2. **Verification**: Uses similarity matching to reduce false positives.
3. **Knowledge Retrieval**: Links violations to specific OSHA regulations.
4. **Automated Reporting**: Generates and emails compliance reports instantly.

System Architecture:



Overall Project Workflow

1. **Ingestion**: User uploads an image via the dashboard
2. **Detection**: Image is tunneled to Colab where **YOLO** identifies PPE status.
3. **Verification**: **CLIP** compares the image against a library of past violations to confirm.
4. **Enrichment**: **RAG** retrieves relevant **OSHA** legal snippets from a local PDF.
5. **Audit**: **CrewAI** agents coordinate to write and verify a professional compliance report.
6. **Notification**: Results are broadcast back to the dashboard and pushed to **n8n** (Docker) for automated email delivery.

AI Processing Pipeline

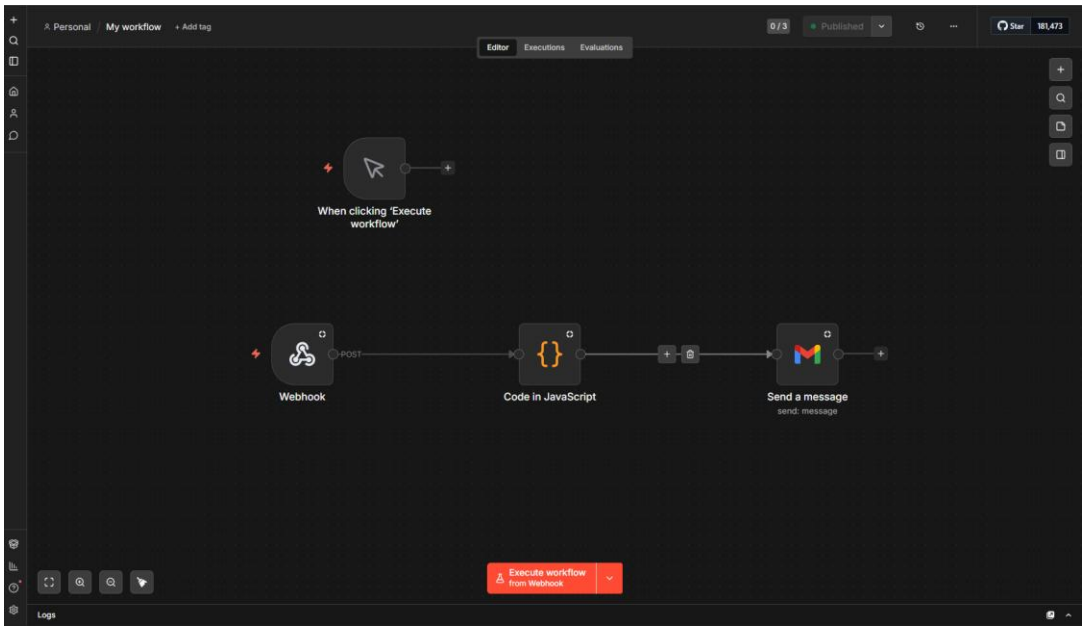
Task	Modality	Model Used	Provider	Rationale
PPE Detection	Computer Vision	YOLOv11	Ultralytics	High-speed real-time detection of helmets, vests, and eyewear
Violation Verification	Computer Vision	CLIP-ViT-B/32	OpenAI (Navigate Labs)	Semantic similarity matching to verify if detected objects truly constitute a breach.
Legal Knowledge Retrieval	NLP (RAG)	GPT-4o-mini	OpenAI (Navigate Labs)	Retrieves hyper-specific OSHA standard snippets for precise legal grounding.
Multi-Agent Orchestration	Autonomous Agents	CrewAI	Multi-Model	Coordinates "Safety Auditor" and "Legal Critic" agents to synthesize reports.
Report Summarization	Text Generation	Gemini 1.5 Flash	Google	Generates concise, human-readable summaries for email notifications.

Connectivity & Hosting Strategy

To overcome infrastructure constraints and ensure accessibility, a multi-tunneling approach is employed:

Component	Hosting	Connection Method	Rationale
Frontend	Local Machine	localhost	Handles sensitive local data and serves the dashboard.
n8n Workflow	Docker Container	ngrok (Static)	Self-hosted for full control; ngrok provides a persistent URL for webhooks.
AI Pipeline & FastAPI Backend	Google Colab	Localtunnel	Uses Colab's T4 GPU for heavy inference; Localtunnel bypasses ngrok free-tier IP limits.

Workflow Orchestration (N8N):



Based on the POC evaluation of the n8n workflow (as seen in the diagram), the orchestration logic is as follows:

1. **Incoming Webhook:** Receives a JSON payload containing the `violation_class`, `agent_report`, and a `base64` encoded image from the FastAPI backend.

2. **Logic Processing (JS)**

- Extracts `image_base64` for email attachment.
- Filters out "safe" detections (STOP logic: `if (violation_class === 'safe') return
- Prepares metadata for the notification.

3. **Communication (Gmail)**: Sends a formatted HTML email titled `Safety Violation Report` to the designated safety officer, including:

- Violation Type & Timestamp.
- The "Legal Critic" reviewed compliance report.
- Attached annotated image showing the breach.

User Experience (UX)

- **Real-Time Dashboard**: Users see live camera feeds or uploaded images annotated with bounding boxes instantly.
- **Instant Alerts**: WebSocket integration ensures the dashboard updates without page refreshes.
- **Automated Audit Trail**: Every violation results in an instant, legally-grounded report, eliminating manual data entry.

Conclusion & Roadmap

The SafetyScanAI POC successfully validates that a hybrid approach—combining local orchestration with cloud-based AI—provides a professional-grade safety solution at a fraction of the cost of dedicated hardware.

Next Steps:

- **Edge Deployment**: Migrating YOLO models to NVIDIA Jetson for on-device inference.
- **Database Integration**: Adding MongoDB or PostgreSQL to store historical violation logs for long-term trend analysis.
- **Multi-Camera Support**: Scaling the FastAPI backend to handle multiple simultaneous streams via RTSP.